
ENGINEERING CLINICS 2024 PROJECT
GROUP - 1

**IMPACT ON DISTRIBUTED NETWORK DUE TO
INTERMITTENT CHARGING OF ELECTRIC VEHICLES AT
IDENTIFIED CLUSTER**

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PROJECT FEATURES

- **Node Identification:** Identifying the optimal node for the EV charging station within the University grid.
- **Peak and off-peak Charging:** Dynamic pricing for EV charging (cheaper during off-peak hours, extra during peak hours). Pricing also depending on the battery type and size, type of charging (Fast or slow)
- **Power Supply Management:** Arrangements to balance the power load to ensure grid stability.
- **Power grid analysis:** Using HOMER Grid software for current power consumption analysis, load simulation, and cost estimation

FLOW DIAGRAM - WORK PROCESS

- **Steps Included:**
 1. Identifying cluster nodes
 2. Power load balancing strategies
 3. Integrating EV Charging infrastructure
 4. Pricing model

FLOW DIAGRAM - VEHICLE CHARGING AT NODE

- **Steps Included:**

1. User plugs in his/her EV
2. System detects peak/off-peak hour
3. Charges are calculated accordingly
4. Power is supplied without overloading the grid

WORKING OF THE PROJECT

- **Analysis of existing power consumption:**
 - ❑ The present power consumption of the University will be analyzed using the HOMER Grid software
- **Node Identification:**
 - ❑ The best node within the University is identified for the Charging Station

- **Power supply management:**

- ❑ During off-peak hours, the power supply management isn't required much, as the aggregate load consumed by the University will still remain within the upper bound of the power supply acquired by the University from the state electricity board.
- ❑ During the peak hours, extra power needs to be supplied to the charging node so as to charge the EV without altering the existing supply load
- ❑ One of the possible solutions is to store the power generated from renewable sources like solar panels, and use that energy to charge the vehicle.

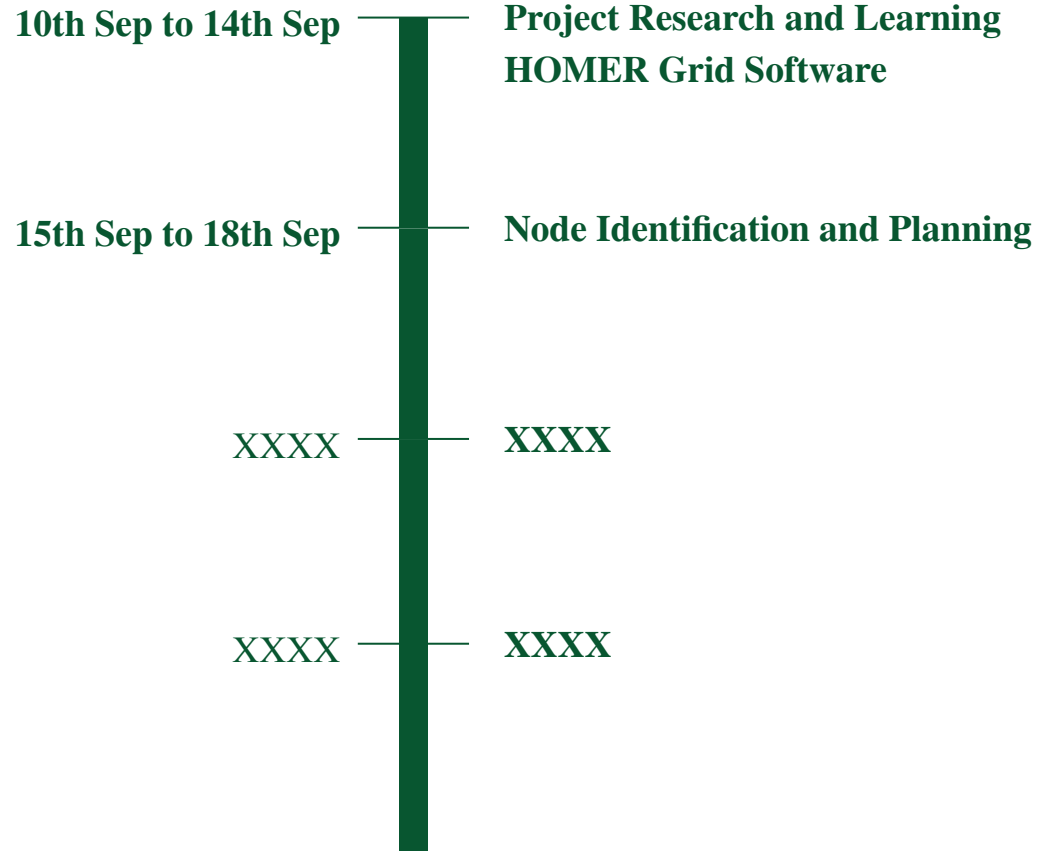
- **Pricing Strategy:**

- ❑ The charging price will be calculated after accounting to various factors, including but not limited to:
 - Battery capacity of the vehicle
 - Type of charging required (Fast/Slow)
 - Peak or off-peak hour
 - No. of vehicles charging at that particular time (demand)

COST ESTIMATION

- HOMER Grid Software Licensing:
 - Student Version - 12 months - \$108.89 (one student)
 - 4 months - \$59.89 (one student)

PROJECT TIMELINE



THANK YOU